

Industrial policy and the creation of the electric vehicles market in China: demand structure, sectoral complementarities and policy coordination

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Since the late 2000s, the Chinese government has been adopting active industrial policies to create a market for electric vehicles. While celebrated as a success nationally and internationally, a closer look reveals a mixed picture with market growth concentrated in only a few cities. On the basis of heterodox industrial policy literature, Chinese-language policy documents and interviews, we develop an analytical framework to empirically study electric vehicles deployment at the city-level, and to assess the achievements and obstacles of implementing industrial policies in this sector. We particularly stress the interrelatedness of policies governing the demand structure of the electric vehicles market and its main complementary sector, the charging infrastructure, which need to be aligned in the progressively more complex segments making up the electric vehicles market. Taking this industry as a case study, we contribute to the wider debate on the determinants of industrial policy effectiveness.

Key words: Industrial policy, China, Electric vehicles, Market creation, Demand structure

JEL classifications: L6; L52; O25

1. Introduction

As part of its efforts to foster industrial development and technological upgrading, the Chinese government promotes the creation of new industries through industrial policy. One of these is the electric vehicles industry, which includes different types of vehicles such as public buses, logistics vehicles and private passenger cars. In China, these became known as New Energy Vehicles (NEVs), as they are powered by different new energy technologies (pure battery electric, plug-in hybrid electric and fuel cell vehicles). Designated as a ‘strategic emerging industry’ in 2010 (State Council, 2010),

Manuscript received 8 October 2020; final version received 30 April 2022

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ambitious targets have been set, including reaching cumulative production and sales of five million electric vehicles by 2020.

By international comparison, the development of the Chinese electric vehicles market appears to be a huge success, with its absolute size exceeding that of other more advanced economies in the 2010s (International Energy Agency, 2019). However, a closer look at the realities on the ground reveals a mixed picture with market growth concentrated in only a few cities. Notably, despite their very similar socioeconomic conditions at the start of China's drive towards NEVs, the two industrial centres of Shenzhen and Suzhou, both among China's wealthiest cities, show highly divergent outcomes in terms of the creation of a market for NEVs. With a total of 480,000 NEVs on the road in 2020 (SHG, 2021), Shenzhen is one of the great successes of China's NEV industrial policy. Suzhou, on the other hand, with only 42,589 NEVs on the road by early 2020 (SZG, 2020), can be considered a laggard given that its economic potential is very similar to Shenzhen's. This article aims to explain these divergent outcomes by providing an in-depth comparative case study of the two cities. By exploring this conundrum, we not only provide a nuanced analysis of industrial policy-making in China, its achievements and obstacles, but also strive to contribute to ongoing debates on the effectiveness of industrial policy in general, and at the technological frontier in particular. Our findings are especially relevant for large middle-income countries characterised by strong state intervention and developmentalism (see Bresser-Pereira, 2020; Nölke *et al.*, 2020).

Drawing on heterodox industrial policy literature, and by triangulating studies on the Chinese NEV market with three distinct data sources—Chinese-language policy documents, Chinese news sources and semi-structured interviews—we develop an analytical framework to empirically study the creation of a market for NEVs at the city level. This framework also helps us to unpack the difficulties of reaching NEV-related targets. It particularly stresses the interrelatedness of policies governing the NEV demand structure and its main complementary sector, the charging infrastructure, which need to be aligned in the progressively more complex segments making up the NEV market. While in simpler NEV segments industrial policy-making is rather straightforward, in more complex segments the mechanisms available for demand management are less direct and the establishment of the required charging infrastructure more difficult. The growing complexity of the tasks results in the involvement of an increasing number of state actors, and policy coordination becomes a key issue. Our in-depth, comparative city-level analysis allows us to take a nuanced view on the implementation of China's industrial policies, unveiling how different explanatory variables interact. We thereby cast doubt on the adequateness of several quantitative studies on the topic which often have contradictory findings.

In the next section, we provide a literature review and identify relevant research gaps. The section also identifies the key NEV policies that kick-started the creation of the Chinese NEV industry. We then present our method, case selection and analytical framework. In the main section, we empirically compare the cases of Suzhou and Shenzhen. In the conclusion, we summarise the findings, outline our contribution to the industrial policy literature and identify the limitations of the study.

2. Literature review and China's electric vehicles policies

We aim to contribute to a long-standing debate about the usefulness and effectiveness of industrial policies in promoting economic development. Typically, arguments

by mainstream economists in favour of industrial policy justify this form of state intervention by pointing to its ability to address various types of market failures, such as imperfect and asymmetric information, knowledge spillovers and other externalities and coordination failures in the intertemporal and intersectoral allocation of resources, which cause the under-provision of certain goods (Hausmann and Rodrik, 2003; Stiglitz *et al.*, 2013; Rodrik, 2014). On the other end of the spectrum, studies highlight the risk of government failures, resulting from governments simply not having the right information about the most promising and economically viable industry to be supported (Pack and Saggi, 2006) and from state capture and rent seeking (Tullock, 2005).

The developmental state literature (e.g., Johnson, 1982; Amsden, 1989; Wade, 1990; Evans, 1995) focuses on the role of the state in promoting industrial transformation and catch-up development through state planning and steering of investment. As noted by Wade (1990, p. 343), while ‘many other nations have at one time or another tried most of the policy tools used in East Asia’, most of them have failed. The successes of East Asian countries were thus predicated on a cohesive state bureaucracy (Evans, 1995) capable of enough autonomy to prevent state capture but simultaneously embedded in the industrial sector in order to build effective state-business alliances (also see Nölke *et al.*, 2020 for an analysis of large middle-income economies such as China, India or Brazil). As states move from technological catch-up to the global frontier of technological development, such as with the NEV industry, they face new challenges—something that the developmental state literature has only recently begun to address (e.g., Kim, 2020). Frontier technologies face higher uncertainty, and thus may require the state to lead the direction of technological change, by shaping and creating new markets.

This problem is *inter alia* addressed in a literature which, similar to the developmental state scholarship, considers the use of industrial policies beyond the mainstream ‘market-fixing’ rationale (Chang and Andreoni, 2016, 2020; Mazzucato, 2016). Building on this literature, Andreoni and Chang (2019) propose that governments should employ multiple policies, simultaneously tackling several issues deemed critical for the successful implementation of industrial policies. These ‘policy packages’, when coherently aligned, should be able to lead the economy in the envisaged direction of industrial transformation.

We highlight three components that are particularly relevant for a policy package to be successfully implemented. The first is demand management: Chang and Andreoni (2016) hold that the literature has often neglected the role of macroeconomic demand in the implementation of industrial policies, and this in turn has led to ‘the neglect of different impacts that changes in demand (sometimes deliberately managed by the government) have on different sectors’ (p. 25). On a macroeconomic level, fiscal, monetary and exchange rate policies can affect aggregate demand. But when it comes to specific sectors—normally the focus of selective industrial policies—demand management must be specifically tailored to the targeted sector. In general, expansion of market demand can enable the realisation of dynamic increasing returns and learning by doing (Arrow, 1962; Kaldor, 1972), fostering supply-side variables such as technological progress. Relatedly, there is a body of literature focusing on the role of demand in fostering technological innovation. Buoyant demand can reduce the uncertainty inherent in the innovation process (Fontana and Guerzoni, 2008) and trigger R&D expenditures by firms (Brouwer and Kleinknecht, 1999). Moreover, Edler and Georghiu (2007) argue that public procurement should be seen as a key component of

demand-oriented innovation. These arguments particularly suit problems identified in the promotion of industrial policies at the technological frontier.

Second, another factor that needs to be taken seriously is the idea of sectoral complementarities, reminiscent of the ‘big push’ argument introduced by [Roseinstein-Rodan \(1957\)](#),¹ in which the viability of an economic activity depends on the existence of another, complementary one. This is essential for industries in which the final demand depends on the existence of a supporting infrastructure—in this case, the NEV charging infrastructure.

Third, since industrial policies entail distributional effects, their effective implementation is dependent on taking the conflicting interests of the agents involved into consideration. Many of these conflicts are found within the state apparatus itself, where ministries, departments and agencies can pursue divergent objectives, leading to a misalignment of policies in the pursuit of industrial transformation ([Andreoni and Chang, 2019](#)). In successful cases of industrial policy implementation, misalignment problems were frequently addressed by shifting the policy process to the upper levels within the governmental apparatus, for example, national, supra-ministerial institutions such as the Ministry of International Trade and Industry in Japan or the Economic Planning Board in South Korea ([Andreoni and Chang, 2019](#); [Nölke et al., 2020](#)). As in many developmental states, China has strong central steering capacity regarding industrial policy (e.g., [Chen and Lees, 2016](#)). However, top-down policy-making here is combined with implementation in variegated local political economies, predicated on China’s institutional organisation (e.g., [McNally, 2012](#); [ten Brink, 2019](#)). Within this context, local governments enjoy strategic agency by implementing national policy guidelines adjusted to local conditions and institutional constraints. As will be shown, these local contexts matter for policy alignment and effective policy implementation.

National electric vehicles policies in China can be traced back to the 1990s, with the focus on basic R&D and the development of prototypes ([Gong et al., 2013](#)). In the 2000s, the national government started to draw up more specific plans and the term NEV eventually evolved to refer to different types of vehicles—private passenger cars, public and private buses, taxis, official vehicles, special purpose vehicles such as garbage and sanitation trucks, and logistics vehicles. The rationale for promoting NEVs is broad and covers different objectives, ranging from reducing oil dependency to achieving technological supremacy in an industry at the technological frontier, as well as mitigating environmental problems (for NEVs as an example of ‘green industrial policy’, see [Altenburg et al., 2017](#)).

In 2009, the national government made a decisive move towards the actual local deployment of NEVs by launching the ‘Ten cities, thousand vehicles’ programme ([Ministry of Finance, 2009](#)). By August 2010, the programme had been expanded to 25 cities ([Gong et al., 2013](#)) including Shenzhen and Suzhou.² The idea was for the central government to provide purchase subsidies and for local governments to fund the charging infrastructure and manage implementation. Initially, the focus was on public service fleets, such as public buses, taxis and government officials’ vehicles. In 2010, the government also introduced measures providing purchase subsidies for private passenger cars ([Ministry of Finance, 2010](#)).

¹ See [Murphy et al. \(1989\)](#) for a mainstream treatment of the ‘big push’ argument.

² By 2016, the programme had expanded to 88 cities in total ([He et al., 2018](#)).

Following these pilots, in 2012, with the ‘Energy-saving and New Energy Automotive Industry Development Programme (2012–2020)’ the [State Council \(2012\)](#) created a national policy that became the foundation for the ‘demonstration and promotion’ of NEVs, including supporting policies such as purchase subsidies, increasingly stringent fuel economy and emission standards for internal combustion engine (ICE) vehicles, and directives for the promotion of the required charging infrastructure ([National Development and Reform Commission, 2015](#)). With these policies, the Chinese government set a cumulative production and sales target of five million by 2020 for NEVs, by which time the technology was expected to be market competitive ([State Council, 2016](#)). With 4.92 million NEVs on the road by the end of 2020, industry analysts consider the target achieved ([He and Jin, 2021](#)).

Combined with the ‘Ten cities, thousand vehicles’ programme, the 2012 programme sets the time frame for our investigation as 2009–20. Towards the end of the decade, purchase subsidies started to be gradually phased out, although they were extended again until the end of 2022 due to the COVID-19 pandemic. Meanwhile new policies targeting carbon emissions of manufacturers’ vehicle fleets and the NEV share of manufacturers’ total production were implemented (the ‘dual credit’ policy, see [Ministry of Industry and Information Technology, 2017, 2020](#)).³

As discussed, policy implementation is largely a local matter in China. The national policy for the promotion of NEVs thus became the foundation for a rapidly growing body of provincial and municipal policies (for an overview, see [Zhang and Qin, 2018](#)). Many quantitative studies attempt to assess the effectiveness of such local NEV policies, but their results are often contradictory. On the one hand, [Ma et al. \(2017\)](#), for instance, conclude that purchase subsidies are effective and should be maintained, and [Deng and Tian \(2020\)](#) hold that, while they are effective in general, they should be differentiated according to the distinct types of electric vehicles in the market. On the other hand, [Qiu et al. \(2019\)](#) argue that purchase subsidies have no significant effect on sales and should be abandoned, and [Sheldon and Dua \(2020\)](#) conclude that high-income consumers in particular could afford electric vehicles even without purchase subsidies, which should therefore be eliminated for this group. This reflects a broader lack of consensus on the effectiveness of Chinese industrial policy in general. Some scholars point to the pivotal role of industrial policies in the country’s industrial upgrading efforts ([Poon, 2009](#); [Lo and Wu, 2014](#); [Chen and Lees, 2016](#)), while others claim that they are ineffective ([Sun, 2007](#); [Chen, 2016](#); [Howell, 2018](#); [Holz, 2019](#)).

We believe that some of these contradictory findings can be explained by taking into account different circumstances of policy implementation at the local level, where many enabling factors interact. However, the literature lacks an in-depth, comparative city-level study, aimed at explaining divergent outcomes regarding NEV deployment. Our research seeks to fill this lacuna. Equally important, NEV studies informed explicitly by an industrial policy framework are also absent. Against this background, Shenzhen and Suzhou represent typical cases for the implementation of the same national policy and as such their implementation strategies can shed light on the effectiveness, or otherwise, of industrial policies in contemporary China and beyond.

³ In October 2020, the State Council released a new industrial development plan covering the period 2021–35, to give sequence to the ‘Energy-saving and New Energy Automotive Industry Development Programme (2012–20)’. The new plan stipulates that by 2025, NEVs sales should reach the figure of 20% of the total annual new vehicles sales.

3. Method and analytical framework

To understand and explain the divergent trajectories of two typical cases, the cities of Shenzhen and Suzhou, we triangulated data from over 100 Chinese-language NEV-related policy documents, both at the national and local levels, Chinese news sources, and 15 semi-structured expert interviews, lasting between one and two hours each (see [Appendices C and D](#) for further information). The latter were conducted with industry and policy experts based in Suzhou, Shenzhen and Beijing, and outside China, in the spring and summer of 2020. Note that due to travel restrictions, the authors, even those resident in China (in this case, Suzhou), had to conduct interviews via on-line communication platforms. Our traditional mixed methods triangulation scheme ([George and Bennett, 2005](#); [Creswell and Plano Clark, 2007](#)) strengthened the reliability of our analysis (for an earlier example of an industrial policy study employing a mixed-method, multidisciplinary research design, see [Andreoni et al., 2017](#)). Our comparative city-level study sheds a light on how different explanatory variables interact in order to elucidate divergent performances in NEVs deployment.

Suzhou and Shenzhen were selected not only because of the important role they have played in China's staggering economic growth in recent decades, but particularly because of their similarities, which enabled us to control for exogenous variables (for an earlier comparison between the two cities in an industrial policy context using a similar research design, see [Chen, 2014](#)).

Shenzhen and Suzhou share similar socioeconomic characteristics. As illustrated in [Table 1](#), at the starting point of our comparison in 2009, the two cities had remarkably similar levels of local GDP, total population, GDP per capita and local budgetary revenue. In that year, with GDP per capita exceeding that of Beijing and Shanghai by almost 25% each, Shenzhen and Suzhou were among China's wealthiest cities ([CEIC Data, 2020](#)).

Moreover, both cities were included in national NEV programmes early on. While Shenzhen was part of the original ten cities programme launched in 2009, Suzhou was included shortly afterwards, in early 2010 ([Howell et al., 2014](#)). When it comes to their industrial structure, both cities are home to a number of important companies that are invested in the NEV industry. While Shenzhen is now famous for being the site of Build Your Dreams (BYD), Suzhou is home to producers of motion batteries, battery management systems, electric engines and high-power transmission systems (such as Lükong), as well as Jinlong-Haige, a leading producer of electric buses. Despite their similar economic potential when China's national drive towards NEVs began, Shenzhen and Suzhou's performance in terms of NEV deployment is highly divergent, a conundrum that we explore using our analytical framework.

Table 1. *Socioeconomic indicators for Suzhou and Shenzhen (2009)*

City	GDP (billion RMB)	Total population (millions)	GDP per capita (RMB/person)	Local budgetary revenue (billion RMB)
Suzhou	774.02	9.37	83,696.00	74.52
Shenzhen	820.13	8.91	87,066.00	88.08

Sources: [Shenzhen Statistical Bureau \(2010\)](#) and [Jiangsu Provincial Bureau of Statistics \(2010\)](#).

3.1 Analytical framework

Focusing on the creation of market demand for NEVs through the implementation of industrial policies, we developed an analytical framework, which emerged from an iterative process involving both inductive theorising from empirical research and deductive theorising from the literature.

The conditions governing the creation of market demand for NEVs can be divided into three interrelated variables: (i) demand structure, (ii) complementary sector and (iii) coordination (see [Figure 1](#)).

The demand structure refers to the composition of market demand, that is, the number of buyers that make up the market and their relative size. Instead of relying on the independent purchase decisions of buyers alone, the state can attempt to manage demand with the objective of boosting NEV deployment. As will be shown, different demand structures have different impacts on the effectiveness of demand management policies.

The main complementary sector is the charging infrastructure sector, without which the demand for NEVs would be seriously undermined. Thus, the state can boost NEV market demand by promoting an adequate charging infrastructure.⁴

Lastly, the policies governing demand conditions for NEVs must be aligned, i.e., they must effectively pursue the same common objective. Notably, where NEV policies are designed, implemented or enforced by different governmental bodies, the chances of misalignment are higher and thus the coordination of policies becomes more difficult.

The NEV label subsumes different segments. Each of these segments features different aspects of the demand structure and the related charging infrastructure. A policy conducive to market demand creation in one particular segment will therefore not necessarily have the same effect in another segment. Consequently, the mechanisms responsible for explaining the development of NEV market demand overall will differ from segment to segment. Our research suggests that the development of the NEV market should be divided into three different segments, featuring the characteristics

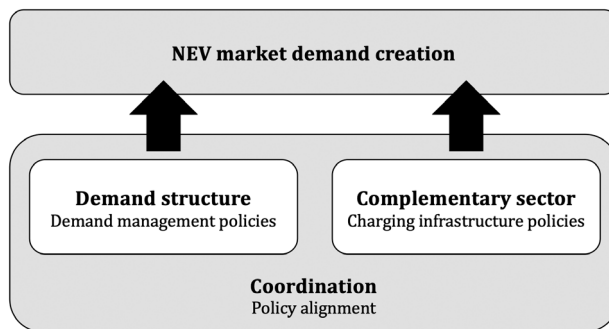


Fig. 1. *NEV market demand creation.*

Source: own elaboration.

⁴ The NEV charging infrastructure is composed mainly by charging stations and charging piles. Charging stations are stations where electric cars are charged with electricity, performing a function similar to gas stations for internal combustion engine (ICE) vehicles. Charging piles are the equivalent to gasoline pumps in gas stations for ICE vehicles. They can be installed in charging stations and also in parking lots or public buildings, and are normally fixed on the ground or walls.

outlined below. The categorisation in Table 2 is thus based on the attributes of our demand-side variables ‘demand structure’ and ‘complementary sector’.⁵

The first segment, *public buses*, is characterised by a single buyer, typically the state-owned local public transport company. Even when cities have more than one public transport company, these are all locally state-controlled companies and subject to the chain of control under the local government. As such, the buyer is within the administrative reach of the local authorities. The required charging infrastructure is characterised by a limited number of charging facilities and their predictable location (normally at initial and end terminals of bus routes and at bus depots).

The second segment, *logistics vehicles*—light and medium trucks, mainly used for short-distance freight delivery in the city—is characterised by a small number of typically private buyers. While the use of logistics vehicles can be fragmented, with multiple small enterprises or individual owners comprising a large share of the market, purchase decisions can be highly concentrated because large companies purchase a sizable number of vehicles and lease them to smaller players. This results in a consolidation of the demand side of the market, making it more responsive to government policies that aim to shape demand for the segment, thus facilitating policy enforcement. The establishment of the charging infrastructure becomes more problematic, however, because the number of charging facilities required is much higher than for public buses

Table 2. *NEV segments: characteristics and main policies*

Segment	Demand structure		Complementary sector	
	Characteristics	Main policies	Characteristics	Main policies
Public buses	Single buyer	- Public procurement - Purchase subsidies	Predictable routes, limited number of vehicles. Need for only small number of stations, at specific locations.	- State-owned enterprises (SOEs) responsible for developing charging infrastructure
Logistics vehicles	Small number of buyers	- Purchase subsidies - Operational subsidies - Regulatory advantages	Less predictable routes, more vehicles. Need for more charging facilities, more widely distributed.	- Private operators become more important - Policy mandates for construction of charging facilities
Private passenger cars	Multiple, decentralised buyers	- Purchase subsidies - Regulatory advantages	Uncertain routes, high number of vehicles. Need for extensive availability of public fast chargers and many home chargers.	- Financial subsidies for charging operators

Source: own elaboration.

⁵ As an analytical device, this categorisation is of course an abstraction of reality, and focuses on the variables (demand structure and charging infrastructure) identified by our literature review and empirical research. Other segments were excluded because either their market is too narrow (e.g., officials’ vehicles) and/or they share hybrid characteristics regarding our variables (e.g., taxis). Their inclusion would not affect our conclusions, however.

and their locations more dispersed. Logistics vehicles tend to recharge by making use of public fast charging stations during the day (Crow *et al.*, 2019) and their promotion therefore provides a more reliable source of demand for charging station operators and assures them a viable return rate. The development of this segment, then, is an important foundation for the advancement of an extensive charging infrastructure network.

The third segment, *private passenger cars*, has a demand structure defined by multiple, decentralised buyers with distinct individual preferences. This multitude of buyers is less responsive to direct government policies aimed at managing the demand for this segment. The establishment of the required charging infrastructure becomes even more problematic, given the higher uncertainty of routes and the sheer number of vehicles. Although the owners prefer to recharge at home, ‘range anxiety’ makes the availability of fast chargers in the city a key concern. Hence, the existence of an extensive network of charging facilities and an actual market for charging operators is of paramount importance.

In general, as Chinese cities advance through these three segments, we find that the available mechanisms for demand management become less direct, the establishment of the required charging infrastructure becomes more difficult and achieving overall policy alignment becomes more burdensome. As a result, market creation for NEVs becomes progressively more challenging.

4. Case comparison

In this section, we empirically compare local, city-level NEV development. We introduce the different NEV policies, analyse their effectiveness and present the reasons why Suzhou did not even come close to the success of Shenzhen. Following our analytical framework, this section is divided into the subsections ‘demand structure’, ‘complementary sector’ and ‘coordination’.

4.1 Demand structure

4.1.1 Public bus segment. Regarding the electrification of public buses, both city governments used national NEV policies to support their local industrial base directly (e.g., SZG, 2014B). Being home to BYD (Shenzhen) and Jinlong-Haige (Suzhou), major bus manufacturers active in national and international markets, unsurprisingly the NEV public bus fleets in both cities are predominantly stocked by these firms. Similarly, the promotion of buses was connected with the status of both cities as ‘National Public Transit Construction Demonstration Cities’, providing them with additional resources for the modernisation of public transport (Ministry of Transport, 2011).

It was relatively easy for the city governments to manage demand in this segment by controlling the only buyer, the local public transport provider. Public procurement was the main mechanism employed to create demand, aided by generous purchase subsidies (Interviews #1 and #4, see Appendix C). By the end of 2017, Shenzhen had deployed a total of 16,359 vehicles, reaching 100% fleet electrification, and becoming the city with the largest electric fleet globally (Lin *et al.*, 2019). Suzhou has also made progress towards fleet electrification, using similar policies. According to the Suzhou Public Transportation Bureau, by mid-2019, of the total of over 5,000 buses, 31% were diesel-electric hybrids, 30% pure battery-electric vehicles and 16% were liquefied

natural gas (LNG) buses, yielding a 77% share of NEVs in the fleet ([Ministry of Transport, 2019](#)).

In summary, both cities exploited the single-buyer demand structure in this segment and their state ownership, managing the demand and shaping the market relatively easily through public procurement and purchase subsidies.

4.1.2 Logistics vehicle segment. While not a frequent subject of discussions, the promotion of electric logistics vehicles (ELVs) proved to be key to Shenzhen's overall success in NEV promotion—unlike in Suzhou where promotional policies were introduced at a later stage. To foster the adoption of ELVs, the Shenzhen city government (SHG) implemented a series of policies culminating in the 'Shenzhen Blue Sustainable Action Plan' ([SHG, 2018](#)). First, new purchases of light duty diesel trucks were prohibited. Second, a number of 'green logistics zones', where only ELVs are allowed to operate, were established in several of Shenzhen's districts. Combined with priority road access, this created a business advantage for owners of ELVs. Alongside these regulatory policies, Shenzhen provided innovative *ex post* operational subsidies for ELVs, granted to owners of fleets of 300 freight vehicles or more with at least 100 ELVs and a minimum mileage accrued annually in the city ([SHG, 2019](#)). These were granted in addition to local purchase subsidies, which were discontinued in mid-2019.⁶

The combination of regulatory, operational and purchase policies boosted Shenzhen's performance. According to [Crow et al. \(2019\)](#), the city managed to leap from 300 vehicles of this type in 2015 to roughly 60,000 by the end of 2018 (around 25% of all NEVs by the end of that year). While direct demand management measures like public procurement are not possible in this segment, local governments can employ more indirect policies—particularly the regulatory policies mentioned above. One interviewee explained: 'really the demand management comes in in the regulation and use of these vehicles... the logistics market is probably something like a middle ground where strong policy tools do exist but not the absolute ability to mandate' (Interview #7, see [Appendix C](#)).

For these policies to be successful, the demand structure of this segment must be made up of a small number of consolidated buyers. Because these are large (private) companies with strong balance sheets, they can afford the high upfront investment costs needed to build a fleet large enough to meet the threshold for receiving operational subsidies. Additionally, they are capable of absorbing short-term costs while they await the *ex post* annual disbursement of subsidies. This was precisely the case in Shenzhen, where three large companies amassed roughly 50% of the market and all met the subsidy eligibility criteria (Interview #7, see [Appendix C](#)). In practice, the policy package employed in Shenzhen altered the relative prices between logistics ICE vehicles and ELVs, favouring the demand for the latter.

In contrast, according to the Suzhou Development and Reform Commission (SZDRC), Suzhou only set itself the very modest target of 721 ELVs for the period 2017–20 ([SZDRC, 2017](#)). The situation may change from 2020 onward, however, because Suzhou has started to implement a set of policies that mirror Shenzhen's initiatives to promote ELVs ([SZG, 2019B](#)). Suzhou now prohibits access to its old town for ICE logistics vehicles, grants operational subsidies for ELVs and seeking to consolidate the demand structure for this segment, requires that all designated pilot firms own at least 100 ELVs to be eligible for operational subsidies ([SZG, 2019C](#)).

⁶ See [Appendix A](#) for further information on operational subsidy standards for ELVs.

Suzhou's adaptation of the policies successfully trialled in Shenzhen is testimony to their effectiveness.

4.1.3 Private passenger car segment. In the absence of more direct public procurement policies and more indirect fleet operational subsidies, local governments focus on other forms of indirect demand management. The most visible of these is the purchase subsidy, with both cities offering almost identical subsidies after 2016. Although the value of these subsidies was significantly reduced by the end of the decade and, following national-level guidance, were phased out by mid-2019, in March 2020 the subsidy program was extended again until 2022 to countervail the negative effects of the COVID-19 pandemic.

In addition to purchase subsidies, indirect regulatory policies have become key for the promotion of private cars in Shenzhen (Interview #4, see [Appendix C](#)): NEVs were allowed to use bus lanes, for instance, as well as roads closed to ICE vehicles ([He et al., 2018](#)). According to the Shenzhen Development and Reform Commission (SDRC), they are also entitled to time-limited free parking ([SDRC, 2015B; 2018A](#)).

The most important regulatory advantage over ICE counterparts came in the form of special conditions for obtaining a new license plate: in order to tackle traffic congestion, Shenzhen introduced a lottery coupled with an auction scheme that new owners of private cars must enter. Winning bids for a new license plate have reached up to 95,000 yuan, a price many cannot afford ([The Economist, 2018](#)). NEV owners are exempt from this scheme, greatly boosting NEVs demand ([D1EV, 2020](#)). Despite a slow start, this policy package eventually bore fruit. In 2014, there were only 4,910 vehicles on the road ([SDRC, 2015A](#)). By 2018, the number had increased to 148,256 vehicles ([Zhu, 2019](#)), reaching 250,000 vehicles in 2020 ([SHG, 2021](#)).

By contrast, as Suzhou has not imposed any system of auction or lottery for the ownership of new cars, NEVs could not be granted an exemption. The same applies to policies regarding road access restrictions, which did not exist until 2019. Given the strong differences in regulatory policies, Suzhou's performance in this segment did not even hit the 30,000-vehicle mark in 2019 ([SZG, 2020](#)).

In summary, the policies implemented in this segment increase the relative prices of ICE vehicles, favouring the demand for NEVs. Contrary to the other segments, however, regulatory policies are even more indirect—the state does not prohibit new purchases of ICE vehicles and cannot employ the type of operational subsidies granted to ELV fleets, given its much less consolidated demand structure. The result is a weaker effect on purchase decisions. Indeed, in 2017 and 2020, Shenzhen's NEVs in this segment amounted to only 42% and 52% of the total number of NEVs, respectively ([SDRC, 2019; SHG, 2021](#)). These numbers contradict the popular perception that Shenzhen is performing exceedingly well in the promotion of private cars—relatively speaking, the city's performance is still reliant on other segments. The regulatory policies also have much clearer limits. They may only work as long as the penetration rate of NEVs in this segment remains low, because as the market becomes dominated by them, there will be a NEV-specific congestion problem, offsetting the policy benefits.

4.2 Complementary sector

The establishment of the complementary sector, the charging infrastructure, has to be developed alongside the promotion of NEVs. In 2012, it became obvious to

the Chinese government that most cities were not managing to provide an adequate network of NEV charging stations and charging piles (Gong *et al.*, 2013). Even in Shenzhen, official documents acknowledged that the city faced problems related to a ‘lagging construction of charging facilities’ (SDRC, 2015A). Likewise, in Suzhou it was recognised that there was an ‘urgent need to speed up the construction of the charging infrastructure’ (SZDRC, 2017, p. 4).

The initial institutional solution for the lack of investment in the sector was to call on state-owned enterprises in the field of electricity supply to take on the responsibility for building charging stations. In Shenzhen, state-owned Putian was instrumental in the establishment of the required charging stations for public buses (Howell *et al.*, 2014), while Suzhou relied on the State Grid’s local subsidiary. Given the limited number of stations required for public buses, as well as their predictable location, this solution worked well.

With the growth of the overall NEV market, however, obstacles to the establishment of the required network of charging facilities became more evident. In particular, securing land for charging facilities is a difficult task in densely populated cities, whether because of price, location or suitability. Charging stations are frequently located away from city centres and business districts, and thus far from the potential demand of daytime NEV users (Interviews #4 and #8, see Appendix C). As a result, NEV users (especially logistics vehicles and taxis) were found queuing up at the most popular charging stations in city centres, while many stations in the suburbs registered very low utilisation rates (Crow *et al.*, 2019; Hove and Sandalow, 2019). In some cases, charging piles were simply abandoned and labelled ‘zombie charging piles’ (Xinhuanet, 2018).

Given these problems, and as additional private charging operators were entering the market, both cities offered financial subsidies and mandated land use policies. Both cities initially subsidised 30% of the construction costs of new charging stations (SZG, 2014A; SDRC, 2015B). After 2015, subsidies started to be disbursed based on the power capacity of the charging stations. At its peak, Shenzhen provided 300 yuan/kWh for slow charging piles and 600 yuan/kWh for fast charging piles, whereas Suzhou offered 600 yuan/kWh for slow charging piles and 900 yuan/kWh for fast charging ones. Regarding policy mandates, both cities specified that all parking spaces in newly constructed buildings should possess the required infrastructure for the construction of charging facilities in the future. Moreover, mandates determined that a minimum percentage of parking spaces in existing and new buildings were to be equipped with charging facilities (SDRC, 2015B; SZDRC, 2017).

In this context, Shenzhen imposed stricter conditions than Suzhou: for example, at least 30% of parking spaces in new public parking lots were required to be equipped with slow charging facilities (SDRC, 2016B); in Suzhou, the corresponding figure is only 10% (SZG, 2016). As a result, the total number of public charging piles in Shenzhen leapt from 3,000 at the end of 2014 (SDRC, 2016A) to roughly 40,000 at the beginning of 2018 (SDRC, 2018B) and more than 80,000 at the end of 2019 (Interview #9, see Appendix C). In contrast, the increase in Suzhou was slower, from 3,909 charging piles at the end of 2016 (SZDRC, 2017) to 11,151 in mid-2019 (SZG, 2019A).⁷

It is telling that, despite more generous financial subsidies, Suzhou still has a much smaller charging infrastructure network. Moreover, it is not only the total number of

⁷ See Appendix B for further information on financial subsidies and land use policy mandates.

charging facilities that matters, but how suitable their location is. This indicates that the most binding constraint in this field is a process of coherent policy planning across the city and its governmental bodies (see Section 4.3). As one interviewee concluded, ‘this is where I would say urban development [planning] and charging infrastructure deployment is something mutually reinforcing’ (Interview #3, see [Appendix C](#)).

4.3 Coordination

For effective policy implementation, policies must be aligned, that is, they should all pursue the same objective. This need to design, implement and enforce policies in a coordinated manner across different governmental bodies points to an essential difference between Suzhou and Shenzhen, as Suzhou is characterised by a more fragmented administrative structure. While both cities govern a number of urban districts (*qu*) in their territory, Suzhou also governs four county-level cities (*xianjishi*): Kunshan, Taicang, Changshu and Zhangjiagang. While districts are fully subordinate to the authority of a city’s prefecture-level government (*dijishi*), county-level cities enjoy more administrative autonomy. To complicate matters, Suzhou’s four county-level cities are all economically powerful and control large territories, creating frictions and hindering administrative coordination ([Cartier, 2016](#)).

One factor that contributes to policy misalignment is the inefficiencies arising from the administrative relationship between prefectures and counties. The transfer of policies to the county-level cities for implementation creates procedural delays and thus policy uncertainty. For example, the local subsidies policy for Kunshan in 2015 was only released in December of that year after already being published by the province (in March) and the prefecture (in August), with subsidies applying retroactively. Earlier that year, a private company in Kunshan had bought electric buses expecting to receive purchase subsidies, only to learn that it did not meet new eligibility criteria when the policy was finally promulgated in Kunshan ([DIEV, 2017](#)). Similarly, while the Suzhou prefecture-level government defines the targets and policies for charging facilities, county-level cities then need to devise their own plans to implement these targets. According to [SZDRC \(2017\)](#), by 2017 only Zhangjiagang ([Zhangjiagang Government \(ZJG\), 2017](#)) and Kunshan ([Kunshan Government \(KSG\), 2017](#)) had actually created such plans for their territories, thus undermining coherent policy planning for charging facilities in Suzhou.

Another factor contributing to misalignment lies in the policy agency of county-level governments. As between other levels of government in China, while county-level cities have to implement upper-level policies according to guidelines, there is potential for strategic agency (or negligence) in their implementation. Additionally, counties may choose to design and implement their own policies. Given that county-level cities often have their own priorities in terms of resource allocation, it is not a given that they will attach the same importance to the achievement of targets as the prefecture (Interview #15, see [Appendix C](#)). For example, Kunshan has established its own NEV Leading Small Group ([KSG, 2020A](#)) and released its own policy directives and targets for NEV promotion ([KSG, 2020B](#)). Tellingly, these documents do not cite any of the documents issued by the Suzhou prefecture-level government.

Kunshan also promoted the use of liquefied natural gas (LNG) buses ([KSG, 2013, 2018](#)), a different technology from electric buses, thus fragmenting demand across

technologies.⁸ Different technologies running in parallel also increase the uncertainty regarding which one will eventually become prevalent, with knock-on effects on the required charging infrastructure, potentially hampering investment by private manufacturers and charging operators.

In contrast, Shenzhen only has urban districts within its administrative structure, which the prefecture steered through the Shenzhen Development and Reform Commission (SDRC) (Interview #1, see [Appendix C](#)). Moreover, Shenzhen also has the special status of a sub-provincial city. On the one hand, this allows it to plan for economic development and to make investment decisions without provincial interference ([Yan and Yuan, 2020](#)). On the other hand, it creates a more direct relationship with the central government, arguably facilitating the implementation of national policies (Interview #15, see [Appendix C](#)). Our document analysis supports this assessment as Shenzhen's NEV policies originate from or are published through the SDRC, whereas in Suzhou issuance is more dispersed among local bodies. In addition, Shenzhen's policy documents do not tend to refer to the provincial level, while Suzhou's do.

Admittedly, Shenzhen also encountered coordination problems. First, when it initially implemented its ELV policies, the 'green logistics zones' were scattered throughout the city. This zone fragmentation led to higher coordination costs when it came to policy enforcement, because it was difficult to monitor all those zones and thus the 'enforcement burden became too high' (Interview #7, see [Appendix C](#)).⁹ The solution envisaged by local officials was to establish one single large zone, thus easing the enforcement burden. Second, Shenzhen's economy is integrated with that of the surrounding cities in the Pearl River Delta, and ELVs often have to travel back and forth between cities. The operational subsidies, however, are only granted when ELVs drive within Shenzhen's perimeter, making it harder to meet the minimum mileage threshold to qualify for them. This represents a disincentive for the purchase of ELVs instead of ICE vehicles, undermining the effectiveness of the policy. This mismatch between a local policy and the existing regional economic integration represents a coordination failure, too (Interview #9, see [Appendix C](#)).

What makes it difficult to resolve these coordination issues is the relative autonomy of China's sub-national governments, and the lack of formal mechanisms to address and align their distinct interests (e.g., [Poncet, 2004](#); [Moore, 2014](#); [Van Aken and Lewis, 2015](#)), problems that have previously been identified as responsible for failures in industrial policy implementation ([Sun, 2007](#); [Chen, 2016](#)). We hold that this sort of coordination issue, predicated on China's central-local institutional arrangements, also helps explain some of the contradictory results found in quantitative studies, which tend to neglect this institutional dimension.

5. Conclusion

Our study assessed the achievements and obstacles of implementing industrial policies in contemporary China, taking an industry at the technological frontier as a case

⁸ A local gas utility company, majority-owned by the county, has been investing in LNG technology since 2008 ([Kunshan China Resources Energy Technology n.d.](#)).

⁹ It is noteworthy how the consolidated demand structure of the ELV segment interacts with the enforcement burden. As one interviewee recounted: 'By consolidating demand, big companies are not going to try to cheat [the subsidies application], the risk is too great' (Interview #7, see [Appendix C](#)). This rationale also explains why it is more difficult to implement operational subsidies for private passenger cars, where the demand structure is decentralised.

study. In summary, by implementing and aligning an array of demand management policies, which in practice shift the relative prices of the industry in favour of electric vehicles, and by imposing stricter policy mandates for the construction of charging facilities, Shenzhen boosted the market demand for electric vehicles, comparatively speaking. Suzhou, in contrast, implemented rather modest policies. In addition, policy alignment and coordination were hampered by the city's fragmented administrative structure, undermining the overall demand for electric vehicles and Suzhou's ability to steer the expansion of the charging infrastructure network. Shenzhen, with its leaner administrative structure, avoided the more severe coordination problems found in Suzhou.

With our study, we contribute, first, to a nuanced analysis of industrial policy-making in China, an arena which has recently become more relevant for international debates on industrial policy. Second, we contribute to the broader industrial policy literature and to the debate on industrial policy effectiveness.

Regarding the first contribution, our analytical framework helped to further a nuanced understanding of NEV policy outcomes. It was able to explain the opportunities and challenges that promoting different NEV market segments entails. In the public buses segment, which is characterised by direct state control over the demand of vehicles, a more limited charging infrastructure and lower coordination costs, the performance of our two cases is quite impressive. Even Suzhou, the laggard in the comparison, performs well here, reflecting the development of this segment in China overall. According to [Xue *et al.* \(2019\)](#), by the end of 2018, around 51% of all public buses in operation in China were pure or hybrid electric vehicles. In 2020, China's fleet accounted for the vast majority of the world's total ([Bloomberg New Energy Foundation, 2021](#)). We also highlighted the importance of logistics vehicles, so far neglected in the literature, as a segment which can substantially shape NEV market development. Let us recall that Shenzhen had more ELVs alone than the total number of NEVs in Suzhou, attesting to the importance of the segment. Recent policies introduced in Suzhou to promote ELVs indicate a growing acknowledgement of its relevance. In 2019, China had roughly 200,000 electric vans and trucks, more than half of the world's total ([Bloomberg New Energy Foundation, 2020](#)).

However, the fact that China wants to promote NEVs in segments in which the state wields only an indirect influence on demand gives rise to difficulties, as the ability of the state to influence the purchasing decisions of agents is weakened. Further, coordination in the establishment of a charging infrastructure and policy implementation is more complex in a process dependent on coherent, city-wide policy planning. Where coordination is lacking, as in the fragmented administration of Suzhou, effective policy implementation is undermined. We also emphasised that cities need to employ policy packages in order to promote NEVs, i.e., there is a need not for one, but for multiple policies, which, if coherently aligned, have every chance of being successful. In this sense, it is difficult to single out whether one specific policy in isolation is effective or not. This could explain the contradictory results found in some econometric studies on the effectiveness of purchase subsidies ([Ma *et al.*, 2017](#) and [Qiu *et al.*, 2019](#)). We suggest that future econometric studies should include whether or not cities' administrative structures are leaner (like Shenzhen) or fragmented (like Suzhou), in the form of a dummy variable.

This then links into the industrial policy literature. Our research confirms the concern that, as [Andreoni and Chang \(2019\)](#) expressed it, without effective coordination different governmental bodies 'might pursue mis-aligned policy initiatives and

interests' (p. 146). Let us recall that for [Wade \(1990\)](#) a key factor explaining why industrial policies were so successful in the East Asian developmental states was the 'consistent and coordinated attentiveness to the problems and opportunities of particular industries, in the context of a long-term perspective on the economy's evolution' (p. 343). The success of China's NEV policies will thus ultimately depend not on one specific policy devised by a smart social planner, but rather on the continual adjustment and coordination of multiple, well-aligned policies across governmental bodies and over time. This presents a real challenge, given the multiplicity of interests and motivations found within China's administrative structures. This still holds true despite the highly centralised and strategic nature of industrial policy-making in the Chinese party-state, even under the Xi Jinping administration.

Additionally, we contribute to a strand of the literature linking the promotion of industrial policies with the role of *domestic* demand ([Butollo and ten Brink, 2018](#); [Jalilian, 2018](#); [Gomes, 2020](#)), especially in a scenario of sluggish external demand. By conducting an in-depth examination of the structure of domestic demand in different market segments, our study reveals some of the limitations a middle-income country like China faces in activating it as an important source of growth. Moreover, as also identified in studies on large middle-income countries (e.g., [Nölke et al., 2020](#); [Allen et al., 2021](#)), we reveal that the potential benefits of China's domestic demand are undermined by the myriad of interests and relative autonomy of its sub-national units.¹⁰ Despite the concerted efforts of the Chinese state, which proved fundamental to the rise of the overall NEV industry and successes in segments such as public buses and ELVs, the very same state faces serious obstacles when implementing industrial policies, even in sectors deemed strategic by policy-makers. Contrary to popular narratives portraying the Chinese state machinery as flawlessly effective, our research has illuminated some of its internal weaknesses.

This article is certainly not without limitations. First, our research could benefit from a more fine-grained analysis of supply-side variables, particularly technological progress. While the focus of the paper was on market demand creation, based on the assumption that, in so far as it creates the market for producers, technological progress would follow suit, future studies may focus on the degree of technological progress and how this affects NEV sales. Second, further research, particularly in middle-income countries, could complement our study by examining whether and to what degree our findings can be generalised beyond China (for an earlier comparison with India, see [Altenburg et al., 2016](#)). Third, more analysis is required to address the environmental impacts of the deployment of NEVs. Despite their purported environmental benefits, there are justifiable concerns about their dependency on coal as a main source of energy, for instance ([International Energy Agency, 2019](#)).

Supplementary data

Supplementary data are available at *Cambridge Journal of Economics* online.

Conflict of interest statement

None declared.

¹⁰ The traditional ICE automobile industry, for example, could be seen as being characterised by a myriad of diverse local interests, imposing higher coordination costs on the central state ([Chu, 2011](#); [Barwick et al., 2017](#)). One interviewee believes that China's biggest mistake was to regard NEV as a distinct industry in relation to traditional ICE automobiles, when, in reality, they have much more in common than policy-makers originally thought (Interview #6, see [Appendix C](#)).

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